

1. RUN CONFIGURATION

Frames Processed	Duration	Processing FPS	Avg Processing	Max Processing
450	1.49 s	301.8	1.75 ms	2.85 ms

2. TRACKING PERFORMANCE

Acquisition Time	Average Tracking Error	Maximum Tracking Error
0.03 s	2.68 px	6.08 px
<= 2 s		<= 10 px
RMSE Tracking Error	Centroid Error Avg/Max	Lock Retention
2.97 px	2.68 / 6.08 px	100.00 %
	ground truth only	
Target Loss	Re-acquisition Avg/Max	Successful Re-acq.
0.00 %	—	0
< 5 %	<= 1 s	0 loss events

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Acquisition Time	0.03	<= 2.0 s	PASS
Tracking Error (Max)	6.08	<= 10 px	PASS
Center Error (Final)	9.22	<= 10 px	PASS
Target Loss	0.00 %	< 5%	PASS
Re-acquisition Time	0.13	<= 1.0 s	PASS
Processing Speed	301.8 FPS	>= 20 FPS	PASS

Ground truth was available: centroid accuracy is measured.

OVERALL: PASS

Source: benchmark video: synthetic_fsoc_beacon.mp4 | Frames: 450 | FPS: 301.8